Runge-Kutta Method OR Runge-Kutta method of 4th order

Consider the differential equation

$$\frac{dy}{dx} = f(x,y) \text{ with initial condition } y(x_0) = y_0$$

The approximate solution of the equation $x_1 = x_0 + h$ is given by

$$y_1 = y_0 + K$$

where $K = \frac{1}{6}[K_1 + 2K_2 + 2K_3 + K_4]$

$$K_1 = hf(x_0, y_0)$$

$$K_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right)$$

$$K_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right)$$

$$K_4 = hf(x_0 + h, y_0 + K_3)$$

Note:- "K" is the weighted mean of K_1, K_2, K_3 and K_4

The generalized R-K method of 4th order formula is given by

$$y_{n+1} = y_n + K$$

where $K = \frac{1}{6}[K_1 + 2K_2 + 2K_3 + K_4]$

$$K_1 = hf(x_n, y_n)$$

$$K_2 = hf\left(x_n + \frac{h}{2}, y_n + \frac{K_1}{2}\right)$$

$$K_3 = hf\left(x_n + \frac{h}{2}, y_n + \frac{K_2}{2}\right)$$

$$K_4 = hf(x_n + h, y_n + K_3)$$

(1) Solve the initial value problem $\frac{dy}{dx} = x+y$, $y(0) = 1$ at the point $x = 0.2$ by using Runge-Kutta method of fourth order by taking $h = 0.2$.

Soln:- Here $f(x,y) = x+y$

$x_0 = 0$, $y_0 = 1$, $h = 0.2$, $x_1 = x_0 + h = 0 + 0.2 = 0.2$

$K_1 = hf(x_0, y_0) = h(x_0 + y_0) = 0.2(0+1) = 0.2$

$K_2 = hf(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2})$

$= h\left[x_0 + \frac{h}{2} + y_0 + \frac{K_1}{2}\right]$

$= 0.2\left[0 + \frac{0.2}{2} + 1 + \frac{0.2}{2}\right]$

$= 0.2[0 + 0.1 + 1 + 0.1]$

$= 0.24$

$K_3 = hf(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2})$

$= h\left[x_0 + \frac{h}{2} + y_0 + \frac{K_2}{2}\right]$

$= 0.2\left[0 + \frac{0.2}{2} + 1 + \frac{0.24}{2}\right]$

$= 0.244$

$K_4 = hf(x_0 + h, y_0 + K_3)$

$= h[x_0 + h + y_0 + K_3]$

$= 0.2[0 + 0.2 + 1 + 0.244]$

$= 0.2888$

$\therefore K = \frac{1}{6}[K_1 + 2K_2 + 2K_3 + K_4]$

$= \frac{1}{6}[0.2 + 0.48 + 0.488 + 0.2888]$

$= 0.2428$

$\therefore y_1 = y_0 + K = 1 + 0.2428$

$= 1.2428$

(2) Given $\frac{dy}{dx} = 3e^x + 2y$ and $y(0) = 0$ find $y(0.1)$ using Runge-Kutta method of fourth order by taking the stepsize $h = 0.1$.

Soln: Here $f(x,y) = 3e^x + 2y$, $x_0 = 0$, $y_0 = 0$, $h = 0.1$

$x_1 = x_0 + h$

$= 0 + 0.1$

$x_1 = 0.1$

$k_1 = h f(x_0, y_0)$ $[\because f(x,y) = 3e^x + 2y]$

$= h[3e^{x_0} + 2y_0]$

$= h[3e^0 + 2(0)]$

$= 0.1(3)$

$= 0.3$

$k_2 = h f(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2})$

$= h[3e^{x_0 + \frac{h}{2}} + 2(y_0 + \frac{k_1}{2})]$

$= 0.1[3e^{h/2} + 2\frac{k_1}{2}]$

$= 0.1[3e^{0.05} + k_1]$

$= 0.1[3 \times 1.0512 + 0.3]$

$= 0.3453$

$k_3 = h f(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2})$

$= h[3e^{x_0 + \frac{h}{2}} + 2(y_0 + \frac{k_2}{2})]$

$= h[3e^{h/2} + 2\frac{k_2}{2}]$

$= 0.1[3e^{0.05} + 0.3453]$

$= 0.3499$

$$k_4 = hf(x_0+h, y_0+k_3)$$

$$= h[3e^{x_0+h} + 2(y_0+k_3)]$$

$$= 0.1[3e^{0.1} + 2(0.3499)]$$

$$= 0.1[3\times 1.1051 + 0.6998]$$

$$= 0.4015$$

$$\therefore k = \frac{1}{6}[k_1 + 2k_2 + 2k_3 + k_4]$$

$$= \frac{1}{6}[0.3 + 0.69076 + 0.69984 + 0.4015]$$

$$= 0.3486$$

$$y_1 = y_0 + k$$

$$= 0 + 0.3486$$

$$= 0.3486$$ is the approximate solution at $y(0.1)$.

(3) using the fourth order Runge Kutta method, Solve the equation $\frac{dy}{dx} = \frac{1}{x+y}$ at the point $x=0.5$, given $y(0.4)=1$ taking $h=0.1$.

Soln:- Here $f(x,y) = \frac{1}{x+y}$, $x_0 = 0.4$, $y_0 = 1$, $h = 0.1$

$$k_1 = hf(x_0,y_0) = h\times\frac{1}{x_0+y_0} = \frac{0.1}{0.4+1} = 0.0714$$

$$k_2 = hf\left(x_0+\frac{h}{2}, y_0+\frac{k_1}{2}\right) = \frac{h}{x_0+\frac{h}{2}+y_0+\frac{k_1}{2}}$$

$$= \frac{0.1}{0.4+0.05+1+0.0357}$$

$$= 0.0673$$

$$k_3 = hf\left(x_0+\frac{h}{2}, y_0+\frac{k_2}{2}\right)$$

$$= \frac{h}{x_0+\frac{h}{2}+y_0+\frac{k_2}{2}}$$

$$= \frac{0.1}{0.4+0.05+1+0.0336}$$

$$= 0.0674$$

$$k_4 = hf(x_0+h, y_0+k_3)$$

$$= \frac{h}{x_0+h+y_0+k_3}$$

$$= \frac{0.1}{0.4+0.1+1+0.0674}$$

$$= 0.0637$$

$$k = \frac{1}{6}[k_1+2k_2+2k_3+k_4]$$

$$= \frac{1}{6}[0.0714+2(0.0673)+2(0.0674)+0.0637]$$

$$= 0.0674$$

$$\therefore y_1 = y_0+k$$

$$= 1+0.0674$$

$$= 1.0674$$

(4) Using the fourth order Runge-Kutta method, solve the initial value problem $y' = x+y^2$, $y(0)=1$ for $x=0.1(0.1)0.2$

at $x=0.1$ and $x=0.2$

Stepsize $h=0.1$

Soln: Here $f(x,y) = x+y^2$, $x_0 = 0$, $y_0 = 1$, $h = 0.1$

$$x_1 = x_0 + h = 0.1$$

$$x_2 = x_1 + h = 0.2$$

I stage:- Solution at x_1

$$k_1 = hf(x_0, y_0) = h(x_0 + y_0^2)$$

$$= 0.1\,(0+1^2)$$

$$= 0.1$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$= h\left[x_0 + \frac{h}{2} + \left(y_0 + \frac{k_1}{2}\right)^2\right]$$

$$= 0.1\,[0 + 0.05 + (1+0.05)^2]$$

$$= 0.1\,[0.05 + (1.05)^2]$$

$$= 0.1152$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$

$$= h\left[x_0 + \frac{h}{2} + \left(y_0 + \frac{k_2}{2}\right)^2\right]$$

$$= 0.1\,[0 + 0.05 + (1+0.05576)^2]$$

$$= 0.1\,[0.05 + 1.1185]$$

$$= 0.1168$$

$$k_4 = hf(x_0 + h, y_0 + k_3)$$

$$= h\,[x_0 + h + (y_0 + k_3)^2]$$

$$= 0.1\,[0.1 + (1+0.1168)^2]$$

$$= 0.1347$$

$$k = \frac{1}{6}[k_1 + 2k_2 + 2k_3 + k_4]$$

$$= \frac{1}{6}[0.1 + 2(0.1152) + 2(0.1168) + 0.1347]$$

$$= 0.1164$$

$\therefore$ Solution at $x_1 = 0.1$ is

$$y_1 = y_0 + k$$
$$= 1 + 0.1164$$
$$= 1.1164$$

II Stage: Solution at $x_2 = 0.2$, $x_1 = 0.1$, $y_1 = 1.1164$, $h = 0.1$

$$k_1 = h f(x_1, y_1)$$
$$= h[x_1 + y_1^2]$$
$$= 0.1[0.1 + (1.1164)^2]$$
$$= 0.1346$$

$$k_2 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{k_1}{2}\right)$$
$$= h\left[x_1 + \frac{h}{2} + \left(y_1 + \frac{k_1}{2}\right)^2\right]$$
$$= 0.1[0.1 + 0.05 + (1.1164 + 0.0673)^2]$$
$$= 0.1551$$

$$k_3 = h f\left(x_1 + \frac{h}{2}, y_1 + \frac{k_2}{2}\right)$$
$$= h\left[x_1 + \frac{h}{2} + \left(y_1 + \frac{k_2}{2}\right)^2\right]$$
$$= 0.1[0.1 + 0.05 + (1.1164 + 0.0775)^2]$$
$$= 0.1575$$

$$K_4 = h f(x_1+h, y_1+k_3)$$

$$= h[x_1+h+(y_1+k_3)^2]$$

$$= 0.1[0.1+0.1+(1.1164+0.1575)^2]$$

$$= 0.1822$$

$$K = \frac{1}{6}[K_1+2K_2+2K_3+K_4]$$

$$= \frac{1}{6}[0.1346+2(0.1551)+2(0.1575)+0.1822]$$

$$= 0.157$$

$$y_2 = y_1+K$$

$$= 1.1164+0.157$$

$$= 1.2734.$$

(5) Employ the Runge-Kutta method to solve $\frac{dy}{dx} = \frac{y^2-x^2}{y^2+x^2}$

$y(0)=1$ at the point $x=0.2$.

Soln: Here $f(x,y) = \frac{y^2-x^2}{y^2+x^2}$, $x_0=0, y_0=1, h=0.2$

$$x_1 = x_0+h = 0+0.2$$

$$x_1 = 0.2$$

$$K_1 = h f(x_0, y_0)$$

$$= h \times \frac{y_0^2-x_0^2}{y_0^2+x_0^2}$$

$$= 0.2\left[\frac{1^2-0^2}{1^2+0^2}\right] = 0.2$$

$$K_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right)$$

$$= h f\left(0 + \frac{0.2}{2}, 1 + \frac{0.2}{2}\right)$$

$$= h f(0.1, 1.1)$$

(x_0, y_0)

$$= 0.2 \times \left[\frac{(1.1)^2 - (0.1)^2}{(1.1)^2 + (0.1)^2}\right]$$

$$= 0.1967$$

$$K_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right)$$

$$= h f\left(0 + \frac{0.2}{2}, 1 + \frac{0.1967}{2}\right)$$

$$= h f(0.1, 1.0983)$$

$$= 0.2 \left[\frac{(1.0983)^2 - (0.1)^2}{(1.0983)^2 + (0.1)^2}\right]$$

$$= 0.1967$$

$$K_4 = h f(x_0 + h, y_0 + K_3)$$

$$= h f(0 + 0.2, 1 + 0.1967)$$

$$= h f(0.2, 1.1967)$$

$$= 0.2 \left[\frac{(1.1967)^2 - (0.2)^2}{(1.1967)^2 + (0.2)^2}\right]$$

$$= 0.1891$$

$$K = \frac{1}{6}\left[K_1 + 2K_2 + 2K_3 + K_4\right]$$

$$= \frac{1}{6}\left[0.2 + 2(0.1967) + 2(0.1967) + 0.1891\right]$$

$$= 0.1959$$

$$y_1 = y_0 + K = 1 + 0.1959$$

$$= 1.1959 \text{ //}$$

(6) By using Runge Kutta method of order 4, Solve the initial value problem $\frac{dy}{dx} - 3x = \frac{y}{2}$ with $y(0) = 1$ at the points $x = 0.1$ and 0.2 with $h = 0.1$.

Soln:- Here $f(x, y) = 3x + \frac{y}{2}$, $x_0 = 0$, $y_0 = 1$, $h = 0.1$

Ans:. $y(0.1) = 1.06653$, $y(0.2) = 1.1672$.

Milne's method

Predictor formula :-

$$y_4^{(P)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$$

Here $f_1 = f(x_1, y_1)$

$f_2 = f(x_2, y_2)$

$f_3 = f(x_3, y_3)$

Milne's Corrector formula

$$y_4^{(C)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(P)})$$

Here $f_4^{(P)} = f(x_4, y_4^{(P)})$

(1) Using the Milne's method, Solve the differential equation $\frac{dy}{dx} = \frac{2y}{x}$, $x \neq 0$ at the point $x=2$, given that $y(1)=2$ $y(1.25) = 3.13$, $y(1.5) = 4.5$ and $y(1.75) = 6.13$.

Soln:- Here $f(x,y) = \frac{2y}{x}$, $h = 0.25$

$x_0 = 1$	$x_1 = 1.25$	$x_2 = 1.5$	$x_3 = 1.75$
$y_0 = 2$	$y_1 = 3.13$	$y_2 = 4.5$	$y_3 = 6.13$

we have to find y_4

we note that $f_1 = f(x_1, y_1) = 2\frac{y_1}{x_1} = 2 \times \frac{3.13}{1.25} = 5.008$

$f_2 = f(x_2, y_2) = 2\frac{y_2}{x_2} = 2 \times \frac{4.5}{1.5} = 6$

$$f_3 = f(x_3, y_3) = 2\frac{y_3}{x_3} = 2 \times \frac{6.13}{1.75} = 7.0057$$

Now, Milne's predictor formula yields

$$y_4^P = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$$

$$= 2 + \frac{4 \times 0.25}{3}(2 \times 5.008 - 6 + 2 \times (7.0057))$$

$$= 2 + 0.3333(10.016 - 6 + 14.0114)$$

$$= 8.0091$$

we note that $f_4^{(P)} = f(x_4, y_4^{(P)}) = 2\frac{y_4^{(P)}}{x_4} = 2 \times \frac{8.0091}{2}$

$$= 8.0091$$

Milne's corrector formula gives

$$y_4^{(C)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(P)})$$

$$= 4.5 + \frac{0.25}{3}[6 + 4(7.0057) + 8.0091]$$

$$= 8.0026$$

(2) Using Milne's method, solve the Differential equation

$\frac{dy}{dx} = x - y^2$, $y(0) = 0$, $y(0.2) = 0.02$, $y(0.4) = 0.0795$

$y(0.6) = 0.1762$ at the point $x = 0.8$.

Soln:- Here $f(x, y) = x - y^2$, $h = 0.2$

$x_0 = 0$	$x_1 = 0.2$	$x_2 = 0.4$	$x_3 = 0.6$
$y_0 = 0$	$y_1 = 0.02$	$y_2 = 0.0795$	$y_3 = 0.1762$

we have to find $y_4 = y(0.8)$

$f_1 = f(x_1, y_1) = x_1 - y_1^2 = 0.2 - (0.02)^2 = 0.1996$

$f_2 = f(x_2, y_2) = x_2 - y_2^2 = 0.4 - (0.0795)^2 = 0.3936$

$f_3 = f(x_3, y_3) = x_3 - y_3^2 = 0.6 - (0.1762)^2 = 0.5689$

By Milne's predictor formula

$$y_4^{(P)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$$

$$= 0 + \frac{4}{3}(0.2)\,(2(0.1996 - 0.3936 + 2(0.5689))$$

$$= 0.3049$$

we find $f_4^{(P)} = f(x_4, y_4^{(P)})$

$$= x_4 - (y_4^{(P)})^2$$

$$= 0.8 - (0.3049)^2$$

$$= 0.7070$$

By Milne's corrector formula, we get

$$y_4^{(C)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(P)})$$

$$= 0.0795 + \frac{0.2}{3}(0.3936 + 4(0.5689) + 0.7070)$$

$$= 0.3045$$

(3) The following table gives the solution of D.Eqn $\frac{dy}{dx} = x^2 + \frac{y}{2}$ for $1 \le x \le 1.3$ at steps of 0.1. Find y at 1.4 using the Milne's predictor-corrector formula.

x	1	1.1	1.2	1.3
y	2	2.2156	2.4649	2.7514

Soln:. Here $f(x,y) = x^2 + \frac{y}{2}$, $h = 0.1$

$x_0 = 1$ $\quad x_1 = 1.1$ $\quad x_2 = 1.2$ $\quad x_3 = 1.3$

$y_0 = 2$ $\quad y_1 = 2.2156$ $\quad y_2 = 2.4649$ $\quad y_3 = 2.7514$

We note that

$$f_1 = f(x_1, y_1) = x_1^2 + \frac{y_1}{2} = (1.1)^2 + 1.1078 = 2.3178$$

$$f_2 = f(x_2, y_2) = x_2^2 + \frac{y_2}{2} = (1.2)^2 + \frac{2.4649}{2} = 2.6724$$

$$f_3 = f(x_3, y_3) = x_3^2 + \frac{y_3}{2} = (1.3)^2 + \frac{2.7514}{2} = 3.0657$$

By Milne's predictor formula

$$y_4^{(P)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$$

$$= 2 + \frac{4(0.1)}{3}(4.6356 - 2.6724 + 6.1314)$$

$$y_4^{(P)} = 3.0792$$

Now $f_4^{(P)} = f(x_4, y_4^{(P)}) = x_4^2 + \frac{y_4^{(P)}}{2} = (1.4)^2 + \frac{3.0792}{2}$

$$= 3.4996$$

By Milne's corrector formula

$$y_4^{(C)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(P)})$$

$$= 2.4649 + \frac{0.1}{3}(2.6724 + 4(3.0657) + 3.4996)$$

$$= 3.0793$$

④ Given $\frac{dy}{dx} = x^2(1+y)$ and $y(1) = 1$, $y(1.1) = 1.233$

$y(1.2) = 1.548$, $y(1.3) = 1.979$, evaluate $y(1.4)$ by Milne's predictor-corrector method.

Soln:. Here $f(x,y) = x^2(1+y)$, $h = 0.1$

$x_0 = 1$ $\quad x_1 = 1.1$ $\quad x_2 = 1.2$ $\quad x_3 = 1.3$

$y_0 = 1$ $\quad y_1 = 1.233$ $\quad y_2 = 1.548$ $\quad y_3 = 1.979$

we have to find $y_4 = y(1.4)$

$$f_1 = f(x_1, y_1) = x_1^2(1+y_1)$$
$$= (1.1)^2(1+(1.233))$$
$$= 2.7019$$

$$f_2 = f(x_2, y_2) = x_2^2(1+y_2)$$
$$= (1.2)^2(1+1.548)$$
$$= 3.6691$$

$$f_3 = f(x_3, y_3) = x_3^2(1+y_3)$$
$$= (1.3)^2(1+1.979)$$
$$= 5.0345$$

By Milne's predictor formula

$$y_4^{(P)} = y_0 + \frac{4h}{3}(2f_1 - f_2 + 2f_3)$$

$$= 1 + \frac{4}{3}(0.1)\left[2(2.7019) - 3.6691 + 2(5.0345)\right]$$

$$y_4^{(P)} = 2.5738$$

we note that $f_4^{(P)} = f(x_4, y_4^{(P)})$

$$= x_4^2 (1 + y_4^{(P)})$$

$$= (1.4)^2 (1 + 2.5738)$$

$$= 7.0046$$

By Milne's corrector formula, we get

$$y_4^{(C)} = y_2 + \frac{h}{3}(f_2 + 4f_3 + f_4^{(P)})$$

$$= 1.548 + \frac{0.1}{3}(3.6691 + 4(5.0345) + 7.0046)$$

$$= 2.5750$$